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Original Research Article

Radiation-dependent demyelination in normal appearing white matter in glioma patients, determined using quantitative magnetic resonance imaging

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ABSTRACT

Background and purpose: A brain tumour, especially a glioma, is a rare disease; it is challenging to treat and the prognosis is often poor. Routine care includes surgery and concomitant chemoradiotherapy (CRT). Diagnostic work-up and treatment effects are typically evaluated using magnetic resonance imaging (MRI). Quantitative MRI (qMRI), unlike conventional MRI, has the advantage of providing tissue-specific relaxation rates and proton density. The purpose is to detect changes in normal appearing white matter (NAWM) in brain tumour patients after CRT using qMRI.

Materials & methods: NAWM was analysed in 10 patients, in 83 MR examinations performed before and after surgery and after CRT. Relaxation rates R_1 and R_2, the proton density (PD) and the concentration of myelin (c_{My}) were calculated from the qMRI scans and analysed in correlation to radiation dose and time after treatment.

Results: A significant decrease in c_{My} between pre-treatment imaging and first follow-up and an increase in PD were observed. For low doses (less than 30 Gy) PD and c_{My} returned to baseline (=pre-operative status), while for high doses (>30 Gy) the change increased during the full extent of the follow-up period. No difference could be established for R_1. For R_2 an increase was observed during the first year, which then gradually returned to baseline. For R_2, stronger effects were seen as a consequence of higher absorbed doses.

Conclusion: In the long-term follow-up for glioma patients, qMRI is a powerful tool for detecting small changes, such as a decrease of myelin concentration, in NAWM after CRT.

1. Introduction

The most common types of brain tumours are gliomas, originating from the glial cells in the brain. Depending on tumour type and malignancy grade, the prognosis varies, with a short overall survival for high-grade gliomas despite an aggressive treatment. After surgery concomitant chemo radiotherapy (CRT) is given with the disadvantage that it affects the surrounding brain tissue as well as the targeted tumour. Also, effects after surgery such as swelling and oedema can cause neurological symptoms and cognitive dysfunctions and these affects are therefore likely to have a long-term impact on quality of life. Damage to the white matter in the brain is often observed within the first year, while radiation effects in the grey matter often do not occur until after six to 12 months [1]. Damage to the grey matter often involves vascular lesions such as telangiectasia and focal haemorrhages, while white matter damage is often transient demyelination [1]. The demyelination is a consequence of the irradiated oligodendrocytes going into apoptosis [2].

For brain tumours and therapy-related affects, MRI is the main imaging technique in clinical practice and includes several images such as T_1W- and T_2W-images, T_2FLAIR, diffusion and perfusion. Assessment of radiological findings is performed according to the RANO criteria ('Response Assessment in Neuro-Oncology Criteria for Gliomas') [3]. To

Abbreviations: MRI, Magnetic resonance imaging; qMRI, quantitative MRI (in this study referred to values obtained by SyMRI-software); ROI, Region of interest; RT, Radiotherapy; CRT, Chemo radiotherapy; 3DCRT, 3D conformal radiotherapy; VMAT, Volumetric modulated arc therapy; C_{My}, Myelin concentration; R_1, Longitudinal relaxation rate, $1/T_1$; R_2, Transverse relaxation rate, $1/T_2$; PD, Proton density.

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Table 1
Characteristics of evaluated patients.

Pat.	Age	Gender	Histopathological diagnosis (WHO 2007)	Dose	Technique	# of follow-up exams
1	63	M	Anaplastic Oligodendroglioma grade III	2 Gy × 30	VMAT	4
2	58	F	Glioblastoma	2 Gy × 30	VMAT	4
3	57	F	Glioblastoma	2 Gy × 30	3DCRT	4
4	61	F	Glioblastoma	2 Gy × 30	VMAT	8
5	65	F	Anaplastic Oligodendroglioma grade III	2 Gy × 30	VMAT	9
6	50	M	Anaplastic Oligodendroglioma grade III	2 Gy × 30	3DCRT	8
7	68	M	Glioblastoma	2 Gy × 30	3DCRT	3
8	43	M	Oligodendroglioma grade II	1,8 Gy × 28	3DCRT	7
9	65	M	Gliosarcoma	2 Gy × 30	VMAT	2
10	45	F	Glioblastoma	2 Gy × 30	VMAT	8

reveal changes and further understand the complicated process in white matter and myelin, diffusion weighted and diffusion tensor imaging has been suggested [4–7]. Another recent study suggested T_2 relaxometry for observing changes after proton radiotherapy [8]. Raschke *et* al examined both these techniques for observing changes in normal appearing white matter (NAWM) [6].

To improve imaging of brain tumours further, quantitative MRI (qMRI) has been proposed, as it provides more detailed and quantitative information about tumour properties. One implementation of qMRI is the QRAPMASTER-sequence together with the SyMRI software [9–11]. This sequence is composed by a combination of interleaved slice selective saturation pulses, and multi echo readouts to generate imaging volumes with a range of echo- and saturation delay- times. These can be used to calculate the qMRI maps; R_1, R_2 and PD ($R_1 = 1/T_1$, $R_2 = 1/T_2$). The qMRI maps can be used for creating synthetic images [12–15]. It can also be used to determine myelin concentrations [16], a method which have been validated using Luxol fast blue staining [17,18]. A major advantage of the technique is the gain of quantitative and tissue specific values, and also the possibility to perform calculations and statistical tests on these. A disadvantage is the variation of quantitative values depending on age and gender [19,20]. QMRI maps are also particularly useful for detecting contrast enhancement in peritumoural oedema [21,22]. Recent published work also concluded that qMRI can be useful for head and neck cancers [23].

The main aim of this project was to investigate if longitudinal qMRI imaging during follow-up after concomitant CRT can be used to detect changes in NAWM. A second aim was to explore if such changes can be related to absorbed dose given in radiotherapy (RT), and if such changes vary over time. The term qMRI will in the following text refer to quantitative values extracted with the QRAPMASTER sequence and SyMRI software.

2. Material and methods

2.1. Subjects

A retrospective analysis of data collected in the period 2013–2018 was performed on a patient cohort consisting of 25 patients with suspected glioma. The characteristics of the peritumoural tissue specifically has been reported previously [21,22]. The data collection was approved by the local ethics committee (Dnr: 2011/406-31) and informed consent was obtained. MR scans were performed before and after surgery and as a follow-up after RT completion every third month. Causes for exclusion were: (i) another diagnosis at histopathology (3), (ii) higher dose per fraction (3.4 Gy × 10) (1), (iii) no follow-up MRI after treatment (4) or (iv) not being treated with RT at our institution (7). After exclusion, ten patients remained for final analysis. The characteristics of the ten analysed patients are shown in Table 1. The histopathological diagnoses were classified according to WHO 2007, which was the standard during the time for inclusion [24].

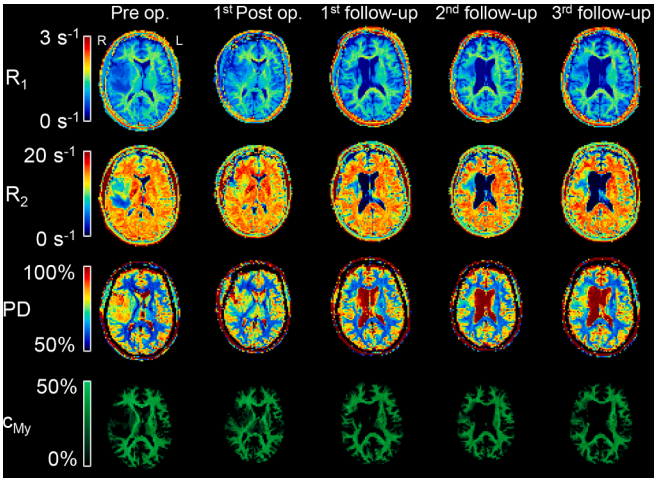


Fig. 1. Example of qMRI-maps for one patient. From top to bottom is the qMRI-maps for R_1, R_2, PD and c_{My} and from left to right is the images from before and after surgery and the three first follow-up images after RT completion, with approximately three months interval.

2.2. Radiotherapy

Nine patients were prescribed the Stupp regime with 60 Gy/30 fractions [25]. One patient, diagnosed with Oligodendroglioma grade II, was treated with 50.4 Gy/28 fractions. The planning target volume, PTV, consists, with margins of 2–3 cm to the cavity from surgery and any residual tumour corrected for anatomical boundaries. Treatment plans were optimized for a mean dose to PTV of 100%, $D_{98\%} \geq 95\%$ and $D_{2\%} \leq 107\%$ although due to proximity of critical organs such as the brainstem not all plans fulfilled the aims. Treatment was delivered using either a TrueBeam or Clinac IX (Varian, Paulo Alto, San Francisco, USA). For volumetric modulated arc therapy (VMAT) 6 MV was used while for 3D conformal plans (3DCRT) a mixture of 6 MV and 15 MV was used, optimized for every patient. Treatment plans were calculated with the AAA algorithm (vers. 10–13) in the treatment planning system Eclipse (vers. 10–13, Varian, Paulo Alto, San Francisco, USA). Dose delivery was verified with phantom QC measurements for VMAT-plans and by measurements with in-vivo diodes during the first treatment for 3DCRT. The department participates annually in remote audits of machine calibration by IROC-H [26].

2.3. MR-acquisition

MR images were acquired on a 3 T Discovery MR750 (GE Medical Systems, Milwaukee, Wisconsin, US) and qMRI images were processed/calculated using SyMRI (vers 0.45.25/27, SyntheticMR AB, Linköping, Sweden). The QRAPMASTER sequence was added to the clinical protocols for brain tumours. The axial scan has a field of view of 220 × 180 mm^2 and a voxel size of 0.43 × 0.43 × 5 mm^3. A total of 24 slices, with 5